NLP tokenization exercises

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Q1 Find the regex that solves table 1.

Match all of these	and none of these
abac	beam
accede	buoy
adead	canjac
babe	chymia
bead	corah
bebed	cupula

Table 1

More regex exercises available at `https://alf.nu/RegexGolf`

🔑 **Q1** `^[a-f]*$`

Q2 Consider the following mini corpus of one sentence:
`start testing stars near streets`

Q2.1 Run the BPE token learner algorithm over the corpus with $k = 3$. What bigram is merged at each step? What is the resulting corpus? What is the vocabulary? To make the exercise less tedious (and because it is not crucial in undersand how the algorithm works), you do not need to add end of word characters.

🔑 **Q2.1**

$k = 1$

merge `(s, t)`

corpus `st a r t`
`t e st i n g`
`st a r s`
`n e a r`
`st r e e t s`

vocabulary `s, t, a, r, e, i, n, g, st`

$k = 2$

merge `(a, r)`

corpus `st ar t`
`t e st i n g`
`st ar s`
`n ea r`
`st r e e t s`

vocabulary `s, t, a, r, e, i, n, g, st, ar`

$k = 3$

merge `(st, ar)`

corpus
```
star t
t e st i n g
star s
n ea r
st r e e t s
```

vocabulary `s, t, a, r, e, i, n, g, st, ar, star`

Q2.2 Run tokenizer inference on the following test corpus:

```
I stare a rare string
```

What is the resulting corpus?

🔑 **Q2.2** Corpus:

```
I
star e
a
r ar e
st r i n g
```